

3. 단순회귀분석

Q3.5) 다음과 같은 회귀직선군 (family of straight lines)

$$E[y_i] = \beta_{0i} + \beta_{1i} x_i, \quad i = 1, 2, \dots, m$$

의 모수들 $(\beta_{01}, \beta_{02}, \dots, \beta_{0m}, \beta_{11}, \beta_{12}, \dots, \beta_{1m}, b_1)$ 들의 다음과 같은을 증명하라.

$$b_1 = \frac{\sum_{i=1}^m \sum_{\mu=1}^{n_i} (x_{i\mu} - \bar{x}_i)(y_{i\mu} - \bar{y}_i)}{\sum_{i=1}^m \sum_{\mu=1}^{n_i} (x_{i\mu} - \bar{x}_i)^2} \quad b_{0i} = \bar{y}_i - b_1 \bar{x}_i, \quad i = 1, 2, \dots, m$$

여기서

$$x_{11}, \dots, x_{1n_1}$$

$$y_{11}, \dots, y_{1n_1}$$

들은 i 번째 직선에 관련된 x_i 와 y_i 들의 측정값이다.

또한 잔차제곱합 (residual sum of squares) SSE와 그의 관련된 자유도 ϕ 가 다음과 같음을 밝히라.

$$SSE = \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \bar{y}_i)^2 - b_1^2 \sum_{i=1}^m \sum_{\mu=1}^{n_i} (x_{i\mu} - \bar{x}_i)^2$$

$$\phi = \sum_{i=1}^m n_i - m - 1$$

[solution]

$$y_i = \beta_{0i} + \beta_{1i} x_i + \varepsilon_i, \quad \varepsilon_i \sim (0, \sigma^2)$$

$$L(\beta_{0i}, \beta_{1i}) = \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \beta_{0i} - \beta_{1i} x_{i\mu})^2$$

$$\frac{\partial L}{\partial \beta_{0i}} = \sum_{\mu=1}^{n_i} -2(y_{i\mu} - \beta_{0i} - \beta_{1i} x_{i\mu})$$

$$\begin{aligned} \frac{\partial L}{\partial \beta_{0i}} \Big|_{\beta = \hat{\beta}} &= \sum_{\mu=1}^{n_i} -2(y_{i\mu} - \hat{\beta}_{0i} - \hat{\beta}_{1i} x_{i\mu}) = 0 \\ &= n_i \bar{y}_i - n_i \hat{\beta}_{0i} - n_i \bar{x}_i \hat{\beta}_{1i} = 0 \quad \bar{y}_i = \frac{1}{n_i} \sum_{\mu=1}^{n_i} y_{i\mu}, \quad \bar{x}_i = \frac{1}{n_i} \sum_{\mu=1}^{n_i} x_{i\mu} \\ \Rightarrow \hat{\beta}_{0i} &= \bar{y}_i - \hat{\beta}_{1i} \bar{x}_i \end{aligned}$$

$$\frac{\partial L}{\partial \beta_{1i}} = \sum_{\mu=1}^{n_i} -2(y_{i\mu} - \beta_{0i} - \beta_{1i} x_{i\mu}) \cdot x_{i\mu}$$

$$\begin{aligned} \frac{\partial L}{\partial \beta_{1i}} \Big|_{\beta = \hat{\beta}} &= \sum_{\mu=1}^{n_i} -2x_{i\mu} (y_{i\mu} - \hat{\beta}_{0i} - \hat{\beta}_{1i} x_{i\mu}) = 0 \\ &= \sum_{\mu=1}^{n_i} x_{i\mu} \cdot y_{i\mu} - \hat{\beta}_{0i} x_{i\mu} - \hat{\beta}_{1i} x_{i\mu}^2 = 0 \\ &= \sum_{\mu=1}^{n_i} x_{i\mu} y_{i\mu} - \{\bar{y}_i - \hat{\beta}_{1i} \bar{x}_i\} x_{i\mu} - \hat{\beta}_{1i} x_{i\mu}^2 = 0 \\ &= \sum_{\mu=1}^{n_i} x_{i\mu} (y_{i\mu} - \bar{y}_i) + \hat{\beta}_{1i} x_{i\mu} (\bar{x}_i - x_{i\mu}) = 0 \end{aligned}$$

$$\therefore \hat{\beta}_{1i} = \frac{\sum_{\mu=1}^{n_i} x_{i\mu} (y_{i\mu} - \bar{y}_i)}{\sum_{\mu=1}^{n_i} x_{i\mu} (\bar{x}_i - x_{i\mu})} = \frac{\sum_{\mu=1}^{n_i} (x_{i\mu} - \bar{x}_i)(y_{i\mu} - \bar{y}_i)}{\sum_{\mu=1}^{n_i} (x_{i\mu} - \bar{x}_i)^2}$$

$$\text{Let } b_{0i} = \hat{\beta}_{0i} \text{ and } b_{1i} = \hat{\beta}_{1i}$$

$$\text{Then, } b_{0i} = \bar{y}_i - b_{1i} \bar{x}_i, \quad (i = 1, 2, \dots, m)$$

$$b_{1i} = \frac{\sum_{\mu=1}^{n_i} (x_{i\mu} - \bar{x}_i)(y_{i\mu} - \bar{y}_i)}{\sum_{\mu=1}^{n_i} (x_{i\mu} - \bar{x}_i)^2}$$

$$\begin{aligned} SSE &= \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \hat{y}_{i\mu})^2 = \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \bar{y}_i + \bar{y}_i - \hat{y}_{i\mu})^2 \\ &= \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \bar{y}_i + \bar{y}_i - \{b_{0i} + b_{1i} x_{i\mu}\})^2 \\ &= \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \bar{y}_i + \bar{y}_i - \{\bar{y}_i - b_{1i} \bar{x}_i + b_{1i} x_{i\mu}\})^2 \\ &= \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \bar{y}_i + b_{1i} (\bar{x}_i - x_{i\mu}))^2 \\ &= \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \bar{y}_i)^2 + (y_{i\mu} - \bar{y}_i) \cdot b_{1i} (\bar{x}_i - x_{i\mu}) \cdot 2 + b_{1i}^2 (\bar{x}_i - x_{i\mu})^2 \\ &= \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \bar{y}_i)^2 - 2b_{1i} \sum_{\mu=1}^{n_i} (y_{i\mu} - \bar{y}_i)(x_{i\mu} - \bar{x}_i) + b_{1i}^2 \sum_{\mu=1}^{n_i} (x_{i\mu} - \bar{x}_i)^2 \\ &= \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \bar{y}_i)^2 - 2b_{1i}^2 \sum_{\mu=1}^{n_i} (\bar{x}_i - x_{i\mu})^2 + b_{1i}^2 \sum_{\mu=1}^{n_i} (\bar{x}_i - x_{i\mu})^2 \\ &= \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \bar{y}_i)^2 - b_{1i}^2 \sum_{\mu=1}^{n_i} (\bar{x}_i - x_{i\mu})^2 = 0 \end{aligned}$$

degrees of freedom: $n - p$

n : number of samples

p : number of estimated parameters

$$n = \sum_{i=1}^m n_i \quad \text{and} \quad p = m + 1$$

$$\therefore \text{df (degrees of freedom)} = \sum_{i=1}^m n_i - (m + 1)$$

+ Show that the Hessian matrix is positive definite

$$\Leftrightarrow \det(A) > 0 \quad \text{and} \quad a_{ii} > 0$$

Q3.6)

다음과 같은 회귀식

$$E[y_i] = \beta_0 + \beta_1 x_i \quad (i = 1, 2, \dots, m)$$

의 모수들 $(\beta_0, \beta_1, \dots, \beta_m)$ 의 최소제곱추정값 $(b_0, b_1, \dots, b_m)$ 들이 다음과 같음을 증명하라.여기서 $x_{i\mu}, y_{i\mu}$ 의 값의 의미는 앞 문제 3-5와 동일하다.

$$b_0 = \frac{\sum_{i=1}^m n_i \left[\bar{y}_i - \hat{x}_i \left(\sum_{\mu=1}^{n_i} x_{i\mu} y_{i\mu} / \sum_{\mu=1}^{n_i} x_{i\mu}^2 \right) \right]}{\sum_{i=1}^m n_i \left(1 - n_i \bar{x}_i^2 / \sum_{\mu=1}^{n_i} x_{i\mu}^2 \right)}$$

$$b_i = \frac{\sum_{i=1}^m \sum_{\mu=1}^{n_i} x_{i\mu} y_{i\mu} - b_0 \sum_{\mu=1}^{n_i} x_{i\mu}}{\sum_{\mu=1}^{n_i} x_{i\mu}^2}, \quad i = 1, 2, \dots, m$$

또한 잔차제곱합은 SSE와 그의 자유도 ϕ_E 가 다음과 같음을 밝히라.

$$SSE = \sum_{i=1}^m \sum_{\mu=1}^{n_i} y_{i\mu}^2 - \sum_{i=1}^m \sum_{\mu=1}^{n_i} x_{i\mu}^2 b_0^2 + b_0^2 \sum_{i=1}^m n_i - 2 b_0 \sum_{i=1}^m n_i \bar{y}_i$$

$$\phi_E = \sum_{i=1}^m n_i - m - 1$$

(Solution)

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, 2, \dots, m$$

$$L(\beta_0, \beta_1, \dots, \beta_m) = \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \beta_0 - \beta_1 x_{i\mu})^2$$

$$\frac{\partial L}{\partial \beta_0} = \sum_{i=1}^m \sum_{\mu=1}^{n_i} -2(y_{i\mu} - \beta_0 - \beta_1 x_{i\mu})_1, \quad \frac{\partial L}{\partial \beta_0} \Big|_{\beta = \hat{\beta}} = \sum_{i=1}^m \sum_{\mu=1}^{n_i} -2(y_{i\mu} - \hat{\beta}_0 - \hat{\beta}_1 x_{i\mu}) = 0$$

$$\sum_{i=1}^m \sum_{\mu=1}^{n_i} y_{i\mu} - \hat{\beta}_0 - \hat{\beta}_1 x_{i\mu} = 0$$

$$\therefore \hat{\beta}_0 = \frac{1}{N} \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \hat{\beta}_1 x_{i\mu}), \quad N = \sum_{i=1}^m n_i$$

$$\frac{\partial L}{\partial \beta_1} = \sum_{i=1}^m \sum_{\mu=1}^{n_i} -2x_{i\mu}(y_{i\mu} - \beta_0 - \beta_1 x_{i\mu}), \quad \frac{\partial L}{\partial \beta_1} \Big|_{\beta = \hat{\beta}} = \sum_{i=1}^m \sum_{\mu=1}^{n_i} -2x_{i\mu}(y_{i\mu} - \hat{\beta}_0 - \hat{\beta}_1 x_{i\mu}) = 0$$

$$\sum_{i=1}^m \left\{ x_{i\mu} y_{i\mu} - x_{i\mu} \hat{\beta}_0 - \hat{\beta}_1 x_{i\mu}^2 \right\} = 0$$

$$\therefore \hat{\beta}_1 = \frac{\sum_{i=1}^m \left(x_{i\mu} y_{i\mu} - x_{i\mu} \hat{\beta}_0 \right)}{\left(\sum_{\mu=1}^{n_i} x_{i\mu}^2 \right)} = \frac{n_i \bar{x}_i \bar{y}_i - n_i \hat{\beta}_0 \bar{x}_i}{n_i \bar{x}_i^2}, \quad \bar{x}_i \bar{y}_i = \frac{1}{n_i} \sum_{\mu=1}^{n_i} x_{i\mu} y_{i\mu}$$

$$\therefore \hat{\beta}_1 = \frac{\bar{x}_i \bar{y}_i - \hat{\beta}_0 \bar{x}_i}{\bar{x}_i^2} \leftarrow \text{insert this into } \hat{\beta}_0 \quad \bar{x}_i^2 = \frac{1}{n_i} \sum_{\mu=1}^{n_i} x_{i\mu}^2$$

$$\hat{\beta}_0 = \frac{1}{N} \sum_{i=1}^m \sum_{\mu=1}^{n_i} \left(y_{i\mu} - \frac{\bar{x}_i \bar{y}_i - \hat{\beta}_0 \bar{x}_i}{\bar{x}_i^2} x_{i\mu} \right)$$

$$\hat{\beta}_0 = \frac{1}{N} \sum_{i=1}^m \sum_{\mu=1}^{n_i} \frac{\hat{\beta}_0 \bar{x}_i}{\bar{x}_i^2} x_{i\mu} = \frac{1}{N} \sum_{i=1}^m \sum_{\mu=1}^{n_i} \left(y_{i\mu} - \frac{\bar{x}_i \bar{y}_i}{\bar{x}_i^2} x_{i\mu} \right)$$

$$\hat{\beta}_0 = \left(1 - \frac{1}{N} \sum_{i=1}^m \sum_{\mu=1}^{n_i} \frac{\bar{x}_i}{\bar{x}_i^2} x_{i\mu} \right) = \frac{1}{N} \sum_{i=1}^m \sum_{\mu=1}^{n_i} \left(y_{i\mu} - \frac{\bar{x}_i \bar{y}_i}{\bar{x}_i^2} x_{i\mu} \right)$$

$$\therefore \hat{\beta}_0 = \frac{\frac{1}{N} \sum_{i=1}^m \sum_{\mu=1}^{n_i} \left(y_{i\mu} - \frac{\bar{x}_i \bar{y}_i}{\bar{x}_i^2} x_{i\mu} \right)}{\left(1 - \frac{1}{N} \sum_{i=1}^m \sum_{\mu=1}^{n_i} \frac{\bar{x}_i}{\bar{x}_i^2} x_{i\mu} \right)} = \frac{\sum_{i=1}^m \left[n_i \bar{y}_i - n_i \bar{x}_i \frac{\bar{x}_i \bar{y}_i}{\bar{x}_i^2} \right]}{N - \sum_{i=1}^m n_i \bar{x}_i^2 / \bar{x}_i^2} = \frac{\sum_{i=1}^m \left\{ n_i \left[\bar{y}_i - \bar{x}_i \frac{\sum_{\mu=1}^{n_i} x_{i\mu} y_{i\mu}}{\sum_{\mu=1}^{n_i} x_{i\mu}^2} \right] \right\}}{\sum_{i=1}^m \left(1 - n_i \bar{x}_i^2 / \sum_{\mu=1}^{n_i} x_{i\mu}^2 \right)} = 0$$

$$\hat{\beta}_1 = \frac{\bar{x}_i \bar{y}_i - \hat{\beta}_0 \bar{x}_i}{\bar{x}_i^2} = \frac{\frac{1}{n_i} \sum_{\mu=1}^{n_i} x_{i\mu} y_{i\mu} - b_0 \frac{1}{n_i} \sum_{\mu=1}^{n_i} x_{i\mu}}{\frac{1}{n_i} \sum_{\mu=1}^{n_i} x_{i\mu}^2} = \frac{\sum_{\mu=1}^{n_i} x_{i\mu} y_{i\mu} - b_0 \sum_{\mu=1}^{n_i} x_{i\mu}}{\sum_{\mu=1}^{n_i} x_{i\mu}^2} \quad 0$$

$$SSE = \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - \hat{y}_{i\mu})^2 = \sum_{i=1}^m \sum_{\mu=1}^{n_i} (y_{i\mu} - b_0 - b_1 x_{i\mu})^2$$

$$= \sum_{i=1}^m \sum_{\mu=1}^{n_i} y_{i\mu}^2 - 2 y_{i\mu} (b_0 + b_1 x_{i\mu}) + (b_0 + b_1 x_{i\mu})^2$$

$$= \sum_{i=1}^m \sum_{\mu=1}^{n_i} y_{i\mu}^2 - 2 \sum_{i=1}^m \sum_{\mu=1}^{n_i} y_{i\mu} (b_0 + b_1 x_{i\mu}) + \sum_{i=1}^m \sum_{\mu=1}^{n_i} (b_0 + b_1 x_{i\mu})^2$$

$$SSE = ① - ② + ③$$

$$② \quad 2 \sum_{i=1}^m \sum_{\mu=1}^{n_i} y_{i\mu} (b_0 + b_1 x_{i\mu}) = 2 \sum_{i=1}^m n_i \bar{y}_i \cdot b_0 + 2 \sum_{i=1}^m b_1 n_i \bar{x}_i \bar{y}_i$$

$$③ \quad \sum_{i=1}^m \sum_{\mu=1}^{n_i} (b_0 + b_1 x_{i\mu})^2 = \sum_{i=1}^m \sum_{\mu=1}^{n_i} b_0^2 + 2 b_0 b_1 x_{i\mu} + b_1^2 x_{i\mu}^2$$

$$= \underbrace{\sum_{i=1}^m \sum_{\mu=1}^{n_i} b_0^2}_{①} + \underbrace{\sum_{i=1}^m b_0 b_1 n_i \bar{x}_i}_{②} + \underbrace{\sum_{i=1}^m b_1^2 n_i \bar{x}_i^2}_{③} \quad ??$$

$$\phi_E = (\text{number of samples}) - (\text{number of estimators})$$

$$= \sum_{i=1}^m n_i - (m+1)$$

Q3.7) 만약

$$\begin{aligned} y_1 &= \beta_0 + \varepsilon_1 \\ y_2 &= 2\beta_0 - \beta_1 + \varepsilon_2 \\ y_3 &= \beta_0 + 2\beta_1 + \varepsilon_3 \end{aligned}$$

이고 $E[\varepsilon_i] = 0$, $i = 1, 2, 3$ 이라면 β_0 와 β_1 의 최소제곱추정값은 무엇인가?

y_i , $i = 1, 2, 3$ 의 함수로써 나타내어라. 그리고 이 경우의 잔차제곱합 (residual sum of squares) 을 구하라.

$$L(\beta_0, \beta_1) = (y_1 - \beta_0)^2 + (y_2 - 2\beta_0 + \beta_1)^2 + (y_3 - \beta_0 - 2\beta_1)^2$$

$$\frac{\partial L}{\partial \beta_0} = -2(y_1 - \beta_0) - 4(y_2 - 2\beta_0 + \beta_1) - 2(y_3 - \beta_0 - 2\beta_1)$$

$$\begin{aligned} \frac{\partial L}{\partial \beta_0} \Big|_{\beta = \hat{\beta}} &= -2(y_1 - \hat{\beta}_0) - 4(y_2 - 2\hat{\beta}_0 + \hat{\beta}_1) - 2(y_3 - \hat{\beta}_0 - 2\hat{\beta}_1) = 0 \\ &= -2(y_1 + 2y_2 + y_3) + \hat{\beta}_0(2 + 8 + 2) = 0 \\ \therefore \hat{\beta}_0 &= \frac{1}{6}(y_1 + 2y_2 + y_3) \end{aligned}$$

$$\frac{\partial L}{\partial \beta_1} = 2(y_2 - 2\beta_0 + \beta_1) - 2(y_3 - \beta_0 - 2\beta_1)$$

$$\begin{aligned} \frac{\partial L}{\partial \beta_1} \Big|_{\beta = \hat{\beta}} &= 2(y_2 - 2\hat{\beta}_0 + \hat{\beta}_1) - 2(y_3 - \hat{\beta}_0 - 2\hat{\beta}_1) = 0 \\ \therefore \hat{\beta}_1 &= \frac{1}{5}(2y_2 - y_3) \end{aligned}$$

$$V^2 L(\beta_0, \beta_1) = \begin{bmatrix} 12 & 0 \\ 0 & 10 \end{bmatrix} = H$$

$$\det(H) = 120 > 0 \text{ and } H_{11} = 12 > 0$$

$\therefore H$ is a positive definite matrix. $\Rightarrow \hat{\beta}_0$ and $\hat{\beta}_1$ are minimum critical points.

$$\begin{aligned} SSE = \sum_{i=1}^3 (y_i - \hat{y}_i)^2 &= (y_1 - \frac{1}{6}(y_1 + 2y_2 + y_3))^2 \\ &\quad + (y_2 - \frac{1}{3}(y_1 + 2y_2 + y_3) + \frac{1}{5}(2y_2 - y_3))^2 \\ &\quad + (y_3 - \frac{1}{6}(y_1 + 2y_2 + y_3) - \frac{2}{5}(2y_2 - y_3))^2 \end{aligned}$$

Q3.8)

두 모수 (parameters) 인 α 와 β 를 추정하기 위하여 다음과 같은 세 가지 형태의 측정값을 할 수 있다.

- (1) α 에 대한 측정
- (2) $\alpha + \beta$ 에 대한 측정
- (3) $\alpha - 2\beta$ 에 대한 측정

위의 모든 측정은 독립이고 기대값이 0이고 분산이 같은 오차를 수반하고 있다.

만약 m 개의 측정값이 (1)에 대하여, 또한 다른 m 개의 측정값이 (2)에 대하여, 그리고 n 개의 측정값이 (3)에 대하여 얻어졌다고 하자. α 와 β 의 최소제곱추정값을 구하고 $m = 2n$ 이 성립하면 α 와 β 의 최소추정값간의 공분산 (Covariance) 이 0임을 증명하라.

- (1) α 에 대한 측정 : $\alpha_1, \dots, \alpha_m$
- (2) $\alpha + \beta$ 에 대한 측정 : $(\alpha + \beta)_1, \dots, (\alpha + \beta)_m$
- (3) $\alpha - 2\beta$ 에 대한 측정 : $(\alpha - 2\beta)_1, \dots, (\alpha - 2\beta)_n$

α 와 β 의 최소제곱추정값 :

$$\alpha_i = \alpha + \varepsilon_i, \quad \varepsilon_i \sim (0, \sigma^2)$$

$$L(\alpha, \beta) = \sum_{i=1}^m (\alpha_i - \alpha)^2 + \sum_{i=1}^m ((\alpha + \beta)_i - \alpha - \beta)^2 + \sum_{i=1}^n ((\alpha - 2\beta)_i - \alpha + 2\beta)^2$$

$$\frac{\partial L}{\partial \alpha} = -2 \sum_{i=1}^m (\alpha_i - \alpha) - 2 \sum_{i=1}^m ((\alpha + \beta)_i - \alpha - \beta) - 2 \sum_{i=1}^n ((\alpha - 2\beta)_i - \alpha + 2\beta)$$

$$\frac{\partial L}{\partial \alpha} \Big|_{\alpha = \hat{\alpha}} = 0, \quad \hat{\alpha} = (\alpha)_1^T \therefore -2 \sum_{i=1}^m (\alpha_i - \hat{\alpha}) - 2 \sum_{i=1}^m ((\alpha + \beta)_i - \hat{\alpha} - \hat{\beta}) - 2 \sum_{i=1}^n ((\alpha - 2\beta)_i - \hat{\alpha} + 2\hat{\beta}) = 0$$

$$\Rightarrow \sum_{i=1}^m \alpha_i + \sum_{i=1}^m ((\alpha + \beta)_i) + \sum_{i=1}^n ((\alpha - 2\beta)_i) - (m+2n)\hat{\alpha} = (2m+n)\hat{\alpha}$$

$$\therefore \hat{\alpha} = \frac{1}{(2m+n)} \left[\sum_{i=1}^m \alpha_i + \sum_{i=1}^m ((\alpha + \beta)_i) + \sum_{i=1}^n ((\alpha - 2\beta)_i) - (m-2n)\hat{\beta} \right]$$

$$\frac{\partial L}{\partial \beta} = -2 \sum_{i=1}^m ((\alpha + \beta)_i - \alpha - \beta) + 4 \sum_{i=1}^n ((\alpha - 2\beta)_i - \alpha + 2\beta)$$

$$\frac{\partial L}{\partial \beta} \Big|_{\alpha = \hat{\alpha}} = 0 \therefore \sum_{i=1}^m ((\alpha + \beta)_i - \hat{\alpha} - \hat{\beta}) - 2 \sum_{i=1}^n ((\alpha - 2\beta)_i - \hat{\alpha} + 2\hat{\beta}) = 0$$

$$\sum_{i=1}^m ((\alpha + \beta)_i) - m\hat{\alpha} - m\hat{\beta} - 2 \sum_{i=1}^n ((\alpha - 2\beta)_i) + 2n\hat{\alpha} - 4n\hat{\beta} = 0$$

$$\therefore \sum_{i=1}^m ((\alpha + \beta)_i) - 2 \sum_{i=1}^n ((\alpha - 2\beta)_i) + (2n-m)\hat{\alpha} = (4n+m)\hat{\beta}$$

$$\therefore \hat{\beta} = \frac{1}{4n+m} \left\{ \sum_{i=1}^m ((\alpha + \beta)_i) - 2 \sum_{i=1}^n ((\alpha - 2\beta)_i) + (2n-m)\hat{\alpha} \right\}$$

insert $\hat{\beta}$ into $\hat{\alpha}(\hat{\beta})$ and rewrite the equation w.r.t. $\hat{\alpha}$

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$$\hat{\alpha} = C_1 + C_2(m-2n)\hat{\beta}, \quad C_1 \text{ and } C_2 \text{ are constants}$$

$$\therefore \text{Cov}(\hat{\alpha}, \hat{\beta}) = C_2(m-2n) \text{Var}(\hat{\beta})$$

$$\text{if } m = 2n, \text{Cov}(\hat{\alpha}, \hat{\beta}) = C_2 \cdot 0 \cdot \text{Var}(\hat{\beta}) = 0$$

No Answers in the textbook.

Gemini 3.1 pro

$$\text{Let } \theta = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$y_{1i} = \alpha + \epsilon_{1i} \quad (i = 1, \dots, m)$$

$$y_{2i} = \alpha + \beta + \epsilon_{2i} \quad (i = 1, \dots, m)$$

$$y_{3i} = \alpha - 2\beta + \epsilon_{3i} \quad (i = 1, \dots, n)$$

Design Matrix X :

$$\begin{bmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \\ 1 & -2 \\ \vdots & \vdots \\ 1 & -2 \end{bmatrix} \left\{ \begin{array}{l} m \times H \\ m \times H \\ n \times H \end{array} \right.$$

$$X^T X = \begin{bmatrix} 2m+n & m-2n \\ m-2n & m+4n \end{bmatrix}$$

$$\begin{aligned} \text{(or)} \quad m \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} + m \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} + n \begin{bmatrix} 1 \\ -2 \end{bmatrix} \begin{bmatrix} 1 & -2 \end{bmatrix} \\ = m \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + m \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + n \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \end{aligned}$$

$$|X^T X| = (2m+n)(m+4n) - (m-2n)^2$$

$$\therefore (X^T X)^{-1} = \frac{1}{(2m+n)(m+4n) - (m-2n)^2} \begin{bmatrix} m+4n & 2n-m \\ 2n-m & 2m+n \end{bmatrix}$$

$$S_1 = \sum_1^m y_{1i}, \quad S_2 = \sum_1^m y_{2i}, \quad S_3 = \sum_1^n y_{3i}$$

$$X^T Y = S_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + S_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + S_3 \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} S_1 + S_2 + S_3 \\ S_2 - 2S_3 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\alpha} \\ \hat{\beta} \end{bmatrix} = (X^T X)^{-1} X^T Y$$

$$= \frac{1}{(2m+n)(m+4n) - (m-2n)^2} \begin{bmatrix} m+4n & 2n-m \\ 2n-m & 2m+n \end{bmatrix} \begin{bmatrix} S_1 + S_2 + S_3 \\ S_2 - 2S_3 \end{bmatrix}$$

(...)

$$\hat{\alpha} = \frac{(m+4n)S_1 + 6nS_2 + 3mS_3}{m^2 + 13mn}, \quad \hat{\beta} = \frac{(2n-m)S_1 + (m+3n)S_2 - 5mS_3}{m^2 + 13mn}$$

$$\text{Cov}(\hat{\alpha}, \hat{\beta}) = \sigma^2 \frac{-(m-2n)}{m^2 + 13mn}$$

$$\therefore \text{if } m = 2n, \text{ Cov}(\hat{\alpha}, \hat{\beta}) = 0$$